

1 Solution — GIAM 2.4

Problem 1

1.1 Problem

Write two-column proofs that verify each of the following logical equivalences. Throughout, t denotes a tautology and c a contradiction.

#	equivalence
1	$A \vee (A \wedge B) \cong A \wedge (A \vee B)$
2	$(A \wedge \neg B) \vee A \cong A$
3	$A \vee B \cong A \vee (\neg A \wedge B)$
4	$\neg(A \vee \neg B) \vee (\neg A \wedge \neg B) \cong \neg A$
5	$A \cong A \wedge ((A \vee \neg B) \vee (A \vee B))$
6	$(A \wedge \neg B) \wedge (\neg A \vee B) \cong c$
7	$A \cong A \wedge (A \vee (A \wedge (B \vee C)))$
8	$\neg(A \wedge B) \wedge \neg(A \wedge C) \cong \neg A \vee (\neg B \wedge \neg C)$

Table 1.1

1.2 The Toolkit

The toolkit. Every justification in the eight proofs below cites one of these. (t = tautology, c = contradiction)

Rearranging commutative $A \wedge B \cong B \wedge A$ $A \vee B \cong B \vee A$ associative $(A \wedge B) \wedge C \cong A \wedge (B \wedge C)$ $(A \vee B) \vee C \cong A \vee (B \vee C)$ distributive $A \wedge (B \vee C) \cong (A \wedge B) \vee (A \wedge C)$ $A \vee (B \wedge C) \cong (A \vee B) \wedge (A \vee C)$ idempotent $A \wedge A \cong A$ $A \vee A \cong A$ double negation $\neg\neg A \cong A$	Moving negations De Morgan $\neg(A \wedge B) \cong \neg A \vee \neg B$ $\neg(A \vee B) \cong \neg A \wedge \neg B$ domination $A \vee t \cong t$ $A \wedge c \cong c$
Collapsing absorption $A \vee (A \wedge B) \cong A$ $A \wedge (A \vee B) \cong A$ complement $A \vee \neg A \cong t$ $A \wedge \neg A \cong c$ identity $A \wedge t \cong A$ $A \vee c \cong A$	What each of the eight collapses to 1. both sides $\cong A$ 2. $\cong A$ 3. both sides $\cong A \vee B$ 4. $\cong \neg A$ 5. $\cong A$ 6. $\cong c$ (a contradiction) 7. $\cong A$ 8. both sides $\cong \neg A \vee (\neg B \wedge \neg C)$
Strategy: start from the messier side and simplify until you reach the other. Every line must be justified by exactly one named law applied to the line above it. <i>If both sides are messy, reduce each separately to a common normal form.</i>	

Figure 1.1: A reference card of laws in three groups. Rearranging: commutative, associative, distributive in both directions, idempotent, and double negation. Collapsing: absorption in both forms, complement giving A or not- A equals t and A and not- A equals c , and identity giving A and t equals A and A or c equals A . Moving negations: De Morgan in both forms, and domination giving A or t equals t and A and c equals c . A summary panel lists what each of the eight equivalences collapses to: 1 both sides equal A ; 2 equals A ; 3 both sides equal A or B ; 4 equals not- A ; 5 equals A ; 6 equals c , a contradiction; 7 equals A ; 8 both sides equal not- A or (not- B and not- C). A closing note gives the strategy: start from the messier side and simplify until you reach the other, justifying every line by exactly one named law applied to the line above.

1.3 The Eight Proofs

Each proof restates the claim, then simplifies one line at a time; every line names the single law that licenses it.

$$1. A \vee (A \wedge B) \cong A \wedge (A \vee B)$$

$$A \vee (A \wedge B)$$

$$\cong A$$

$$\cong A \wedge (A \vee B)$$

given

absorption

absorption, read right-to-left

Both sides collapse to A , so they are equivalent to each other.

$$\mathbf{2.} \quad (A \wedge \neg B) \vee A \cong A$$

$$\begin{array}{ll} (A \wedge \neg B) \vee A & \text{given} \\ \cong A \vee (A \wedge \neg B) & \text{commutative } (\vee) \\ \cong A & \text{absorption, with } \neg B \text{ for } B \end{array}$$

Proof 3 starts from the messier right-hand side.

$$\mathbf{3.} \quad A \vee B \cong A \vee (\neg A \wedge B)$$

$$\begin{array}{ll} A \vee (\neg A \wedge B) & \text{given} \\ \cong (A \vee \neg A) \wedge (A \vee B) & \text{distributive } (\vee \text{ over } \wedge) \\ \cong t \wedge (A \vee B) & \text{complement} \\ \cong A \vee B & \text{identity} \end{array}$$

$$\mathbf{4.} \quad \neg(A \vee \neg B) \vee (\neg A \wedge \neg B) \cong \neg A$$

$$\begin{array}{ll} \neg(A \vee \neg B) \vee (\neg A \wedge \neg B) & \text{given} \\ \cong (\neg A \wedge \neg \neg B) \vee (\neg A \wedge \neg B) & \text{De Morgan} \\ \cong (\neg A \wedge B) \vee (\neg A \wedge \neg B) & \text{double negation} \\ \cong \neg A \wedge (B \vee \neg B) & \text{distributive, factoring out } \neg A \\ \cong \neg A \wedge t & \text{complement} \\ \cong \neg A & \text{identity} \end{array}$$

$$5. A \cong A \wedge ((A \vee \neg B) \vee (A \vee B))$$

$$\begin{aligned}
& A \wedge ((A \vee \neg B) \vee (A \vee B)) && \text{given} \\
\cong & A \wedge (A \vee \neg B \vee A \vee B) && \text{associative } (\vee) \\
\cong & A \wedge (A \vee A \vee (\neg B \vee B)) && \text{commutative, associative} \\
\cong & A \wedge (A \vee (\neg B \vee B)) && \text{idempotent} \\
\cong & A \wedge (A \vee t) && \text{complement} \\
\cong & A \wedge t && \text{domination} \\
\cong & A && \text{identity}
\end{aligned}$$

$$6. (A \wedge \neg B) \wedge (\neg A \vee B) \cong c$$

$$\begin{aligned}
& (A \wedge \neg B) \wedge (\neg A \vee B) && \text{given} \\
\cong & (A \wedge \neg B) \wedge (\neg A \vee \neg \neg B) && \text{double negation} \\
\cong & (A \wedge \neg B) \wedge \neg(A \wedge \neg B) && \text{De Morgan, read right-to-left} \\
\cong & c && \text{complement}
\end{aligned}$$

$$7. A \cong A \wedge (A \vee (A \wedge (B \vee C)))$$

$$\begin{aligned}
& A \wedge (A \vee (A \wedge (B \vee C))) && \text{given} \\
\cong & A \wedge A && \text{absorption, with } B \vee C \text{ for } B \\
\cong & A && \text{idempotent}
\end{aligned}$$

$$8. \neg(A \wedge B) \wedge \neg(A \wedge C) \cong \neg A \vee (\neg B \wedge \neg C)$$

$$\begin{aligned}
& \neg(A \wedge B) \wedge \neg(A \wedge C) && \text{given} \\
\cong & (\neg A \vee \neg B) \wedge (\neg A \vee \neg C) && \text{De Morgan, twice} \\
\cong & \neg A \vee (\neg B \wedge \neg C) && \text{distributive, factoring out } \neg A
\end{aligned}$$

1.4 Remark — The Two Proof Shapes

The eight proofs use only two overall shapes, and recognising which applies saves a great deal of flailing.

Shape 1 — one side is already simple. In 2, 4, 5, 6, 7, and 8, one side is a bare letter (or c , or a short expression), so the proof is a straight-line simplification of the *other*

side down to it. Always start from the messier expression: simplifying is easier than complicating, because at each step you can see what to cancel.

Shape 2 — both sides are messy. In 1 and 3 neither side is obviously simpler. The technique is to reduce **each side separately** to a common **normal form** and let transitivity of $\cong$ close the gap. In proof 1 both sides collapse to A ; the middle line is that shared form, and the second half of the proof is the first half’s reasoning run backwards. This is the habit noted in Problem 2.3.2.

1.5 Remark — Proof 6: Spotting a Disguised Complement

Proof 6 is the cleverest of the eight. The expression is

$$\underbrace{(A \wedge \neg B)}_X \wedge (\neg A \vee B),$$

and the second factor is exactly $\neg X$ in disguise: by De Morgan, $\neg(A \wedge \neg B) \cong \neg A \vee \neg\neg B \cong \neg A \vee B$. So the whole thing is $X \wedge \neg X$, which the complement law sends to c immediately. Without noticing the disguise one is tempted to distribute everything into four terms and cancel them individually — which works, but takes four times as long. **Looking for a sub-expression and its negation** is a habit worth building; it is also what made Problem 2.2.7 short.

1.6 Remark — Absorption Does Most of the Work

Proofs 1, 2, 5, and 7 all turn on **absorption**, $A \vee (A \wedge B) \cong A$ and $A \wedge (A \vee B) \cong A$. Proofs 5 and 7 in particular are near-instant once you see the shape $A \wedge (A \vee \textit{anything})$: whatever fills the slot is irrelevant, so the whole expression is just A . In proof 5 the “anything” is $\neg B \vee (A \vee B)$; in proof 7 it is $A \wedge (B \vee C)$. Both could be done in a single line by citing absorption directly — the longer versions above are given to show the mechanism, but recognising the shape at a glance is the real skill.

1.7 Remark — Why Two-Column Proofs Are Worth the Trouble

A truth table would verify all eight of these mechanically (Problem 2.3.2 did exactly that for a similar list). So why bother with the laws? Three reasons. First, **scale**: proofs 7 and 8 have three variables and so need $2^3 = 8$ rows; a formula with ten variables needs **1024**, and the algebraic route does not grow that way. Second, **discovery**: a truth table can *confirm* an equivalence you already suspect, but it cannot *produce* a simpler form — the laws can, which is what a compiler or circuit optimiser actually needs. Third, **generality**: the two-column format forces every step to name its licence, so the same discipline transfers directly to the theorem-proving of Chapter 3, where no truth table is available at all.

1.8 Remark — On the Notation t and c

The symbols t and c stand for a **tautology** (always true) and a **contradiction** (always false), and they behave like the constants **1** and **0** of ordinary algebra:

logic	arithmetic analogue
$A \wedge t \cong A$	$x \cdot 1 = x$
$A \vee c \cong A$	$x + 0 = x$
$A \wedge c \cong c$	$x \cdot 0 = 0$
$A \vee t \cong t$	<i>(no analogue)</i>

Table 1.2

The last row is where the analogy breaks: nothing in arithmetic swallows everything the way t does under $\vee$. That extra collapsing power is what makes proofs like 5 possible, and it is the same asymmetry noted in Problem 2.3.1 — Boolean algebra is not arithmetic with new symbols.